SIDE-EFFECT TERM MATCHING FOR COMPUTATIONAL ADVERSE DRUG REACTION PREDICTIONS

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Objectives

The establishment of polypharmacological networks, all the interactions between a collection of drugs and proteins, will allow for the exploration of drug re-purposing, side effect prediction, and the development of more efficacious drugs, targeting multiple proteins in a disease pathway. This project will help pave the way for the computational prediction of adverse drug reactions using a polypharmacological network built using molecular docking scores, an efficient prediction of how and how well drugs bind to a protein. The research presented here is an effort to map side-effect terms associated with a toxicity screen[1], known proteins in which drugs should not interact, to known side-effects associated with FDA-approved drugs[2] in order to test the accuracy of predictions.

Significance

This significant research will help alleviate the current economic burden of developing new pharmaceuticals by innovatively utilizing massive computational power. This research will lead to the establishment of structures to use in virtual drug screenings that will predict side-effects quickly and efficiently, resulting in safer clinical trials, as fewer drugs with negative off-target effects will advance to this stage, and more affordable therapies, as drugs destined for failure will be predicted earlier in the drug discovery process. This project will address important public health concerns by providing safer and more affordable drugs.

Process

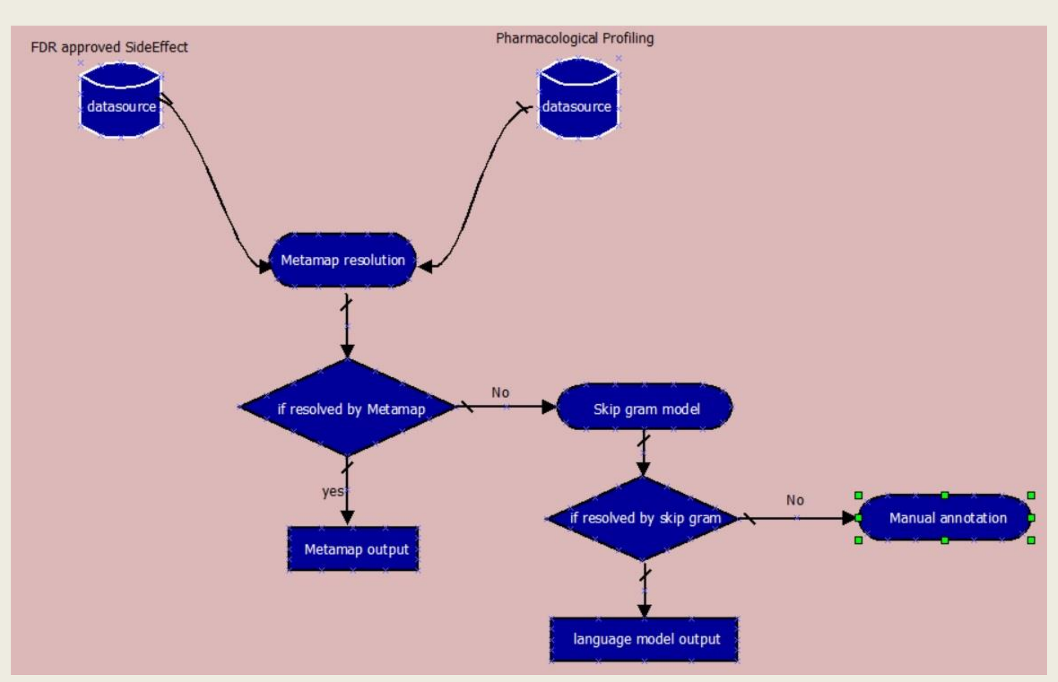
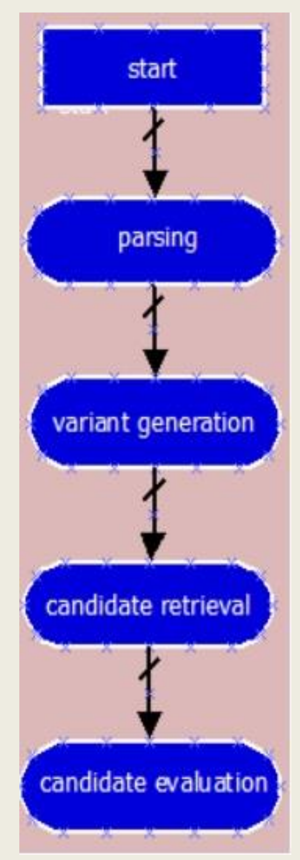


Fig1: Flow Diagram

Methods

Multiple methods were tested for term matching.

1. Edit Distance determines the minimum amount of editing required to transform a source string into a destination string. Editing operation includes insertion, deletion and substitution. Each operation involves a cost and the goal is to find the minimum cost path.
2. Knuth Prat Morris is a pattern-matching algorithm that finds the number of times a given text pattern occurs within a text document, which has been modified to include a cost function that will measure how much the pattern is matched with given text. This algorithm maintain a prefix function for a given pattern which is actually a finite state machine to encapsulate the knowledge about how the pattern matches against the shift of itself.
3. Sense Disambiguation using meta map takes advantage of UMLS (Unified medical language system) metathesaurus, which is a large multi purpose thesaurus containing medical and health related terms, lexicons and group them into concepts. In order to generate final result it goes through following four steps,



Step1 Parsing: text parsed in noun phrases using xerox POS tagger to perform syntactic analysis.

Step2 Variant Generation: Varaints for each input phrase are generated using the knowledge of specialist lexicons and supplementary database of synonyms

ocular {[adj], 0= ""}
eye {[noun], 2="s"}
eyes {[noun], 3="si"}
optic {[adj], 4="ss"}
ophthalmic {[adj], 4="ss"}
ophthalmia {[noun], 7="ssd"}
oculus {[noun], 3="d"}
oculi {[noun], 4="di"}

Fig2 : Variants for word ocular

Step3 Candidate Retrieval: the candidate sets retrieved from the UMLS metathesaurus contains at least One of the variants generated from step three. The amount of matched candidate sets can be controlled By pre setting some parameter value.

Step 4 Candidate Evaluation: Each of the retrieved candidate set is evaluated against the input text. First the Input text is mapped to one of the candidate word then the strength of the mapping is calculated by using a Cost function that involve the calculation of four different metrics: centrality(involvement of head word),variation (an average of inverse distance scores), coverage(how much the candidate word matches input text) and Cohesiveness(how many pieces matching occurs).

861 complications <1> (Complication)
861 complications <3> (Complications Specific to Antepartum or Postpartum)
777 Complicated
694 Ocular
638 Eye
638 Eye NEC
611 Ophthalmic
611 Optic (Optics)
588 Ophthalmia (Endophthalmitis)

Fig 3: Metathesaurus candidates for phrase Ocular complications

Leftmost score represents the metamap calculated phrase mapping score. Names in bracket will be generated If the candidate name does not represent the preferred name as it is generated in the top two candidates of Fig3.

4. Language Model: Distributed representation of words in n dimensional vector space has very interesting impact on natural language processing tasks. Vector representation of words explicitly encode many linguistic regularities and patterns..

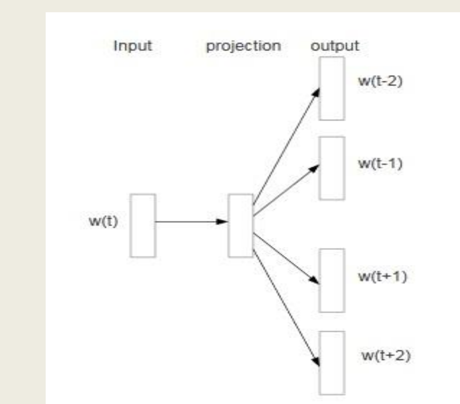


Fig4: Skip gram model [3]

Mathematically for a given set of words w_1, w_2, w_3 the target to maximize the following average distribution,

$$\frac{1}{T}\sum_{t=1}^{T}\sum_{-c\le j\le c, j!=0}\log p(w_{t+j}|w_t)$$

here c is the size of the context, larger c means more training data and better prediction result. The basic Skip gram formulation using softmax function is,

$$P(w_o|w_i) = \frac{e^{(v\prime_{w_o} v_{w_i})}}{\sum_{w=1}^{W} e^{(v\prime_{w} v_{w_i})}}$$

Final result will have the following distributional effect.

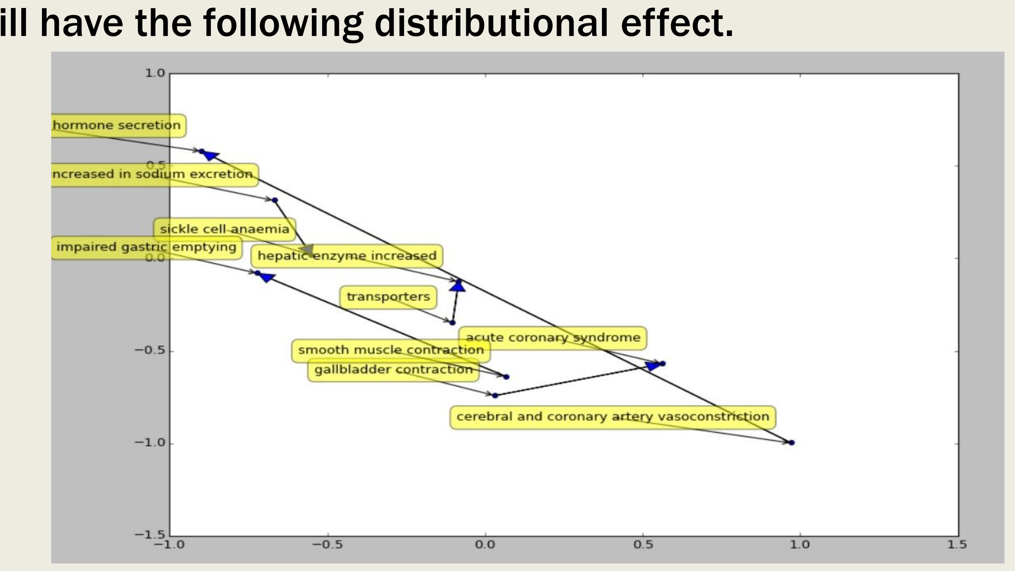


Fig6: distributed mapping

Results

Sense disambiguation appears to be the most accurate methods but only resolved around 75% of the terms. Multi-approach methods are being investigated to achieve the maximum number of terms matched with a high-degree of accuracy.

Vitro Profiled side effect name	Matched FDR side effect name	Preferred Name	Concept cui
Potential cardiac valvulopathy	cardiac valve disease	heart valve disease	c0018824
cardiac hypertrophy	cardiac hypertrophy	cardiac hypertrophy	c1383860
in body weight	weight decreased	body weight	c0005910

Table I: matching result for meta map

Vitro Profiled side effect name	Matched FDR side effect name	Max score	Min score
cerebral and coronary artery vasoconstriction	Coronary artery disease	8.33754030863	-1.0
gastric emptying	Impaired gastric emptying	6.83925867081	-1.0
gallbladder contraction	Gallbladder disorder	9.01812928915	-1.0

Table II: matching result for Skip Gram

References

1. Bowes, J.; Brown, A. J.; Hamon, J.; Jarolimek, W.; Sridhar, A.; Waldron, G.; Whitebread, S., Reducing safety-related drug attrition: the use of in vitro pharmacological profiling. Nature Reviews Drug Discovery 2012, 11 (12), 909-922.
2. Kuhn, M.; Letunic, I.; Jensen, L. J.; Bork, P., The SIDER database of drugs and side effects. Nucleic acids research 2015, gkv1075.
3. Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg Corrado, Jeffrey Dean Distributed representation of words, phrases and their compositionality

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